

UNIVERSITY OF WATERLOO, ON, Canada

WMO: 713685

Lat: 43.4735N

Lon: 80.5572W

Elev: 334

StdP: 97.38

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.4	-23.4	0.5	-20.3	-19.4	0.7	-17.2	7.0	-2.9	6.3	-2.3	0.5	320	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.5	23.3	29.0	22.5	27.5	21.6	25.1	28.3	23.9	27.2	22.9	26.1	1.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.9	27.3	22.9	18.4	25.8	21.9	17.2	24.8	78.5	28.3	73.5	27.3	69.5	26.1	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.2	4.4	-25.5	32.8	3.8	2.1	-28.2	34.3	-30.4	35.5	-32.6	36.7	-35.3	38.2		
			-26.4	26.6	4.3	1.9	-29.5	28.0	-32.0	29.1	-34.5	30.2	-37.6	31.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-6.3	-5.7	-0.6	6.3	13.1	18.3	20.7	19.8	16.3	9.4	3.3	-2.8
	DBStd	10.76	6.46	6.01	6.17	4.97	4.73	3.90	3.11	2.99	4.08	4.63	4.61	5.42
	HDD10.0	2104	504	439	333	132	21	1	0	0	2	69	206	398
	HDD18.3	4113	762	673	586	362	176	48	12	19	88	279	452	657
	CDD10.0	1270	0	0	6	20	116	249	331	304	191	50	4	0
	CDD18.3	237	0	0	0	1	13	47	84	65	26	2	0	0
	CDH23.3	2017	0	0	2	8	128	418	733	493	219	16	0	0
	CDH26.7	493	0	0	0	0	24	110	198	108	51	2	0	0
(14) Wind	WSAvg	1.6	2.0	2.0	2.1	2.2	1.8	1.3	1.1	1.1	1.1	1.4	1.7	1.8
(15) Precipitation	PrecAvg	967	86	66	65	80	91	82	99	69	85	79	87	75
	PrecMax	1189	129	107	113	141	162	174	175	142	228	156	162	140
	PrecMin	679	45	31	32	33	39	42	15	15	26	24	38	41
	PrecStd	131	27	25	26	28	30	31	40	28	48	36	33	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.8	21.1	24.6	29.4	31.2	32.7	31.8	30.6	25.4	17.6	12.8
		MCWB	10.6	9.8	17.2	17.5	21.5	23.5	25.4	23.8	23.0	19.7	14.0	11.7
	2%	DB	7.4	7.5	15.5	20.8	26.7	29.6	30.7	29.3	28.1	21.9	14.7	9.2
		MCWB	6.9	6.3	11.8	15.0	20.0	22.3	23.9	22.8	21.7	17.6	12.6	8.3
	5%	DB	4.3	4.6	12.0	18.1	24.5	27.8	28.8	27.6	25.9	19.4	12.8	6.5
		MCWB	3.6	3.1	9.5	12.8	18.0	21.6	22.6	22.2	20.5	16.4	11.0	5.6
	10%	DB	2.1	2.1	8.6	15.2	21.9	25.8	27.2	26.1	23.7	17.0	10.7	4.0
		MCWB	1.6	1.3	6.4	10.7	16.2	20.4	21.6	21.3	19.4	14.7	9.2	3.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.9	17.5	19.1	23.7	25.4	27.5	26.1	25.1	21.4	14.9	11.8
		MCDB	10.8	11.4	20.7	21.9	27.1	28.8	31.5	28.2	28.5	23.9	16.3	12.2
	2%	WB	7.1	6.8	12.8	16.4	21.1	23.7	25.5	24.4	23.2	18.8	13.1	8.6
		MCDB	7.3	7.8	14.9	19.7	24.7	27.6	28.7	27.6	26.5	20.7	14.4	9.1
	5%	WB	3.7	3.3	9.8	14.1	19.3	22.6	24.1	23.2	21.6	16.8	11.1	5.7
		MCDB	4.2	4.5	11.8	16.9	22.8	26.3	27.0	26.3	24.3	18.9	12.5	6.3
	10%	WB	1.6	1.4	6.5	11.3	17.3	21.5	22.8	22.2	20.3	15.0	9.3	3.4
		MCDB	2.0	2.2	8.4	14.5	21.0	24.7	25.9	24.9	22.8	16.8	10.6	4.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	9.2	10.5	11.0	10.6	11.0	10.7	11.2	9.3	7.5	6.5
		MCDBR	8.8	10.6	13.9	15.6	13.5	12.6	12.7	12.1	13.7	12.8	11.4	8.5
	5% WB	MCWBR	8.5	9.4	10.3	10.4	7.9	6.8	6.7	6.6	7.8	8.7	9.2	7.8
		MCWBR	8.5	10.1	13.1	13.3	12.1	11.0	10.9	10.3	11.2	11.5	10.4	8.4
(40) Clear-Sky Solar Irradiance	taub		0.286	0.331	0.325	0.360	0.386	0.417	0.409	0.397	0.368	0.351	0.305	0.284
		taud	2.366	2.182	2.389	2.335	2.306	2.235	2.278	2.297	2.374	2.409	2.502	2.437
	Ebn at Noon		846	848	916	904	886	856	858	857	858	825	824	817
		Edh at Noon	78	112	105	120	127	137	130	123	105	90	69	67
(44) All-Sky Solar Radiation	RadAvg		1.40	2.25	3.52	4.45	5.39	5.96	5.92	5.17	4.14	2.48	1.52	1.00
	RadStd		0.19	0.29	0.37	0.50	0.51	0.39	0.40	0.26	0.40	0.24	0.21	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page